The artificial hand with elastic fingers for humanoid robot ROMAN

Mianowski K.
Warsaw University of Technology
Warsaw, Poland
e-mail: kmianowski@meil.pw.edu.pl,

Berns K., Hirth J.,
Robotics Research Lab, Dept. of Computer Sciences
University of Kaiserslautern
Kaiserslautern, Germany
j_hirth, berns@informatik.uni-kl.de

***Abstract*— The original design of multifingered gripper in the shape of human hand with flexible fingers dedicated for humanoid robot ROMAN is presented in the paper. The design work was concentated on ensuring that the hand can contact with a variety of different shape objects made of flexible materials of different properties as well as fragile objects requiring particular caution when manipulated. Additionally it was assumed that the gripper should ensure a satisfactory reliabilty level, required high functionality and easy maintenance. The project was based on the assumptions, that the final solution should characterized by high versatility, good adaptability and good robustness. Design methods and prototyping of a robotic hand is presented in the paper in details. In the real solution of the model, the plastic and composite materials were applied. In control systems, PIC controllers and PC computers have been used.**

I. INTRODUCTION

Certain basic problems connected with reproduction of function of lively organisms in design and construction of robots undertakes biomechanics and cybernetics. By exploiting very simple biocybernetic models, it is possible to create mechanical devices similar to human, not only from geometrical point of view, but some functional characteristics too. Such a models can imitate behaviour and living functions of a human body, arms, hands, legs, head, and by applying artificial receptors, can extract information from the environment for executing advisable exercices similarly to the human. Modern robotics takes up new challeges to ensure that the robots are capable of executing more and more difficult tasks and manipulators with grippers can perform a class of new prehension functions. Nowadays, service robotics is one of most interesting fields of robot applications, involving the desire for substituting for a human being in incomplicated manipulation processes that require precise object identification, gentle grasping and precise object handling for manipulation. There arises the need for the application of special grippers provided with sensing capability. To ensure that the gripper could execute manipulation tasks typical of human being to perform, it seems resonable to use the one with antropomorphic shape. A human hand has 5 fingers and up to 23 degrees of freedom, while the whole hand with the arm and forearm has 30 degrees of freedom [5]. Acording to the literature data available the minimal number of fingers necessary for successful performance of the prehension, i.e., the process consisting in grasping the object with a simultaneous leaning against the palm surface is equal to three (four-point grasp) [1,5]. Each human finger has 3 to 4 degrees of freedom, therefore, requiring proper control. Additionally, the control of fingers should ensure their proper cooperation, corresponding to the object shape. Therefore, on the one hand a solution to the grasping problem should reveal adequate generality level, and on the other hand, a general approach should be able to perform using a number of specific, relatively simple methods. Primary analysis of the grasping process proves that application of three-finger-grasp is most profitable. The design work was concentated on ensuring that the hand can contact with a variety of different shape objects made of flexible materials of different properties as well as fragile objects requiring particular caution when manipulated. Additionally it was assumed that the gripper should ensure a satisfactory reliabilty level, required high functionality and easy maintenance. The gripper design ensures beneficial variability of characteristics of forces acting upon the object irrespective of the openning distance of fingers i.e., of the manipulated object size. When manipulating different objects in the course of a service task a human being uses two arms, and his/her hand are most frequently engaged. For such a task, typical of human work should be executed by a robot, it requires appropriate adaptation of both the manipulator robot kinematical structure and grasping devices as well as working environment in the way ensuring that a robot not only can replace a human being in specified manual operations but also can execute the task faster, with better accuracy and more precise. Therefore, the system should reveal a high level of adaptability and be provided with a number of sensing devices of different types allowing for gathering the information on environment that is necessary for manipulation performance. Modern 3D CAD systems allow for a general approach to the design process including determination of basics

978-1-4673-5508-7/13/$31.00 ©2013 IEEE

characteristics of the designed systems and the integration of subsystems. Therefore high qualities of the design (like, high functionality and aesthetics) can be achieved as early as at the design stage. At the initial stage of the developed design of multi-fingered gripper it was assumed that a gripper consisted of a thumb (with additional revolution relative the longitudinal axis in its staring position) and four fingers with redundant DOFs pressing against the object. The fingers were actuated by means of the flexible elements that ensured automatic control of the pressure exerted by particular phalanges upon the object. The driving systems were situated in the gripper body, while the phalanges were driven through connecting rods provided with additional systems of elastic springs. By adding flexibility to the fingers, we improve substantially the gripper functionality through rising its level of anthropomorphism. The fingers will be covered with a special flexible glove with embedded contact pressure force sensors and other measurement gauges. The gripper was designed for grasping an object by means of hooking clamping, grasping, supporting - stabilized by the other hand/gripper (like in a typical two-hand manipulation). Additionally, the design was supplied with the following means:

- force sensors, pressure sensors, contact sensors,
- touch sensor for identification the presence of an object
- image processing methods for identification of geometry of the manipulated object to achieve optimal prehension and grasping processes as well as optimal control of the manipulation task execution.

In the developed design to ensure highest performance of the system is was assumed that all mechanical, electric and electronic elements (necessary driving systems, gear and transmission systems, joints, sensors, etc.) will be integrated with the mechanical system of gripper . The mechanical part was designed using the ProEngineer system and then the initial simulations were performed, allowing for estimation of basics characteristics and their optimisation according to the assumed criteria. A general scheme of the gripped of the properties described above, that performs a grasping prehension is shown in Figure 1.

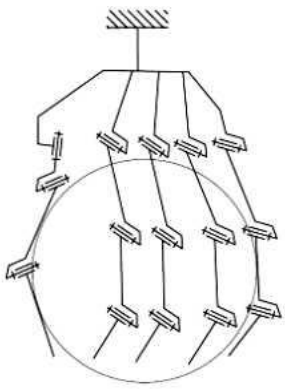
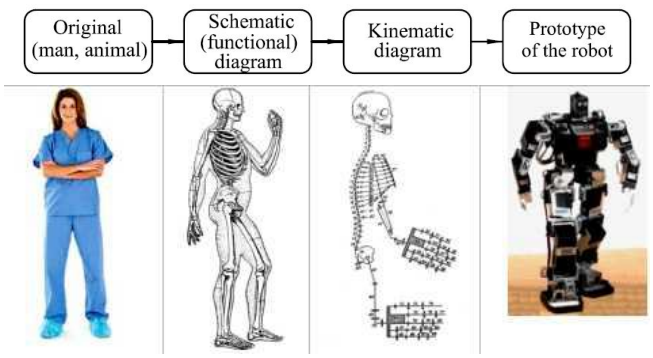


Figure 1. Basic scheme of the gripper performing a grasping prehension.

To simplify the control process when manipulating the object it was assumed that the performance of one finger is subject to kinematic control and it plays a the *leading* role, while the other fingers are situated on the other side of the manipulated object and they stabilize the object through the flexible shape adaptation to the remaining surfaces of an object. The one presented in Fig. 1 on the left plays the above role. It is a simple plane mechanism (2D) with 3 DOFs and consists of two phalanges.In the course of grasping process the finger has to execute the task of supporting the gripped object at three points situated in the plane of its motion, grasping it at the same time in the way preventing it from slipping out. Hereinafter, the finger will be considered as the so-called *leading* one. Four the other fingers are similar mechanisms, but having an additional degree of freedom, i.e., rotation. They are also capable of grasping object but in contrast to the *leading* finger their working planes may be turned and adapted for cooperations purposes with the object to allow for grasping it from three directions. In a general case, the grasping task formulated in the above way should ensure that each phalange contacts with the object, that would require a separate drive for each phalange as well as measurement of the contact force acting upon the object at many points. Since it was planned that driving systems provided with additional sensors to measure contact forces would be situated in the gripper base a design of a single finger following the above conept with transmission systems might occur extremely complex. Additionally, the determination of current configuration of the object-gripper system might be difficult; namely, for such a system with rigid links the rigid problem of dynamics (kinetostatics) is statically undefined. When attempting at control of such a gripper one may be confronted with obstacles resulting in damage of the manipulated object or gripper failure.

II. BASIC IDEA OF THE MECHATRONIC DESIGN

In Fig. 2. the general scheme of the design process is shown. Basis on the nature (Fig. 2a), anatomical atlas represents functional parts of the living creature i.e. bones (Fig. 2b), muscles, internal organs etc. From this representations, the designer can to describe kinematic digram (Fig. 2c), in which kinematic relations are shown, and number of degrees of freedom is defined. Appropriately to aimed target the prototype of the robot can be designed, as it is shown in Fig. 2d.

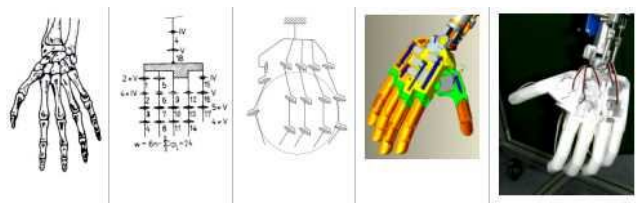


a) b) c) d)

Figure 2. General scheme of imitation of the nature:
a) original (man), b) functional diagram, c) kinematic diagram, d) prototype of the robot.

A proper grasping of the object, that frequently depends on many factors is of crucial importance in the prehension. A precise control of the co-operation between fingers requires the prior determination of accurate kinematical and dynamical models of each finger and the updated information on parameters of particular parts of kinematical chain of the gripper. In particular, the measurements of forces are affected by non-linear friction components, mostly of indetermined nature, while the force measurement performed by means of surface sensors suffer from large errors due to noises and other influences in the signal processing channels. Therefore, in the design process special attention was focused on choosing the mechanical system structure that allowed for minimisation of measurement errors arising from its work.

Started from the sketch presented in anatomical atlas (Fig. 3a), the kinematic diagram (Fig. 3b) and functional kinematic diagram of the hand was created as it it shown in Fig. 3c. This kinematic diagram of the hand with five fingers has 15 degrees of freedom. It is still too many DOF's to drive them by the motors located inside the palm, and so we have analysed basic functions of each finger. Real fingers are driven by elastic tendons, and the basic function of the hand is to catch 3D objects. We have decided to transmit motions with using connecting rods with built-in elastic springs.



a) skeleton of human hand | b) basic kinamatic diagram | c) simplified kinamatic diagram | d) 3D model of the hand | e) prototype of the gripper of human hand type

Figure 3. Scheme how to create human-like artificial robot hand in the base of the theory of cybernetics.

In a mechanical part of the gripper each given kinematical task may be executed by mechanisms of different types; like, cams, levers, rods, toothed mechanisms or mixed. Therefore some types of simple mechanisms were chosen that allow for minimisation of both mechanical noises and mechanical hysteresis from small motors with gears located inside the palm with additional kinematic coupling between following phalanges of each finger and with applying elastic springs in the transmission system. Because ot this, kinematic diagram shown in Fig. 3c need six motors only – two motors for the thumb, and four motors for the other fingers - one motor for each finger. In the design process of gripper, the following phases were considered:

- choice of the mechanism type;
- selection of a structure within that type;
- determination of geometrical parameters.,

with special attention focused on minimisation of the undesirable effects mentioned above. That required the ball bearing to be itroduced into the design which additionally complicated the solution.

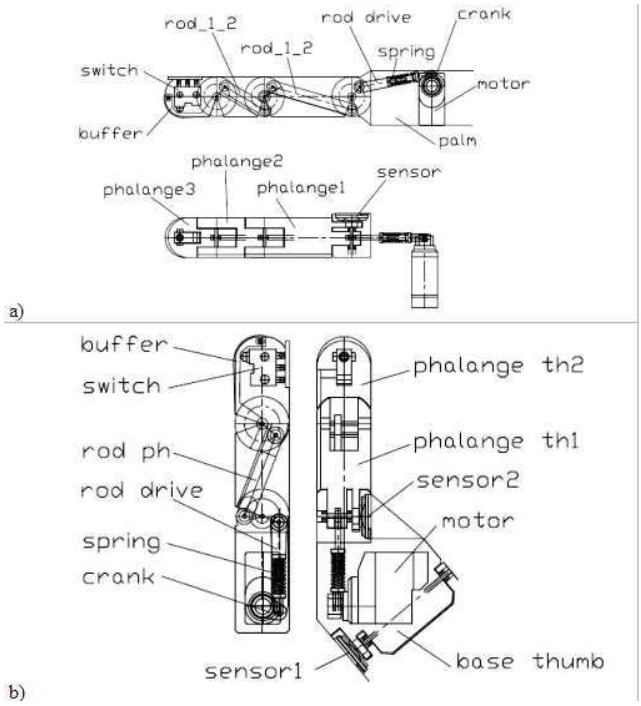


Figure 4. Schematic diagram of the driving system of:
a) four main fingers with kinematic coupling of phalanges motion,
b) thumb with kinemtic coupling of phalanges.

Kinematics of all the hand is presented in Fig. 5.

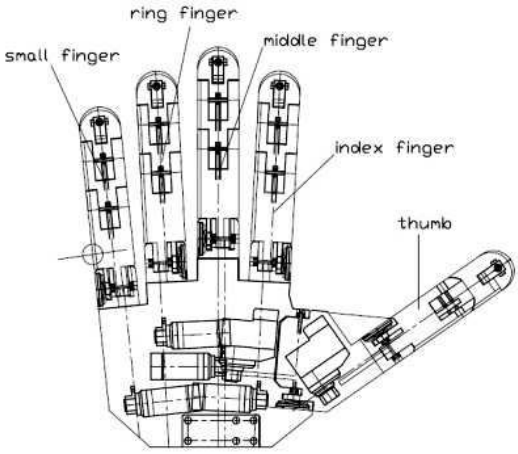


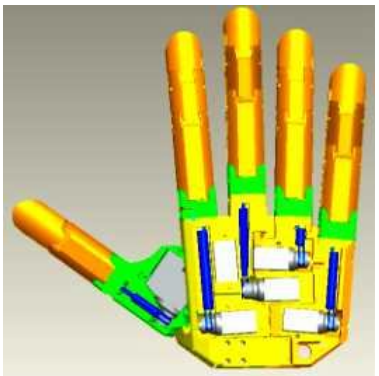
Figure 5. Schematic diagram of the designed hand.

Functional kinematic diagram of the finger is shown in Fig. 4a. In proposed solution special kinematic coupling between phalanges in each finger was applied (the crossed five bar). Last and medium phalanges during the revolution of the first phalange, rotates realised actual absolute angle, which value is an angle of previous phalange plus the value of angle of first phalange. By appropriate control, the hand can realize the motion in form outwards/inwards - opening/closing the hand. Each finger exerts the pressure upon the object in the direction

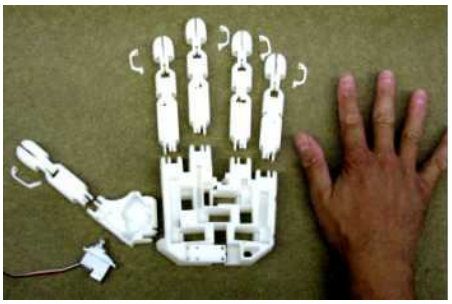
perpendicular to the internal surface of phalange, the clamping force is produced by elastic force from rod driven by appropriate motor.

III. PROTOTYPING OF THE HAND

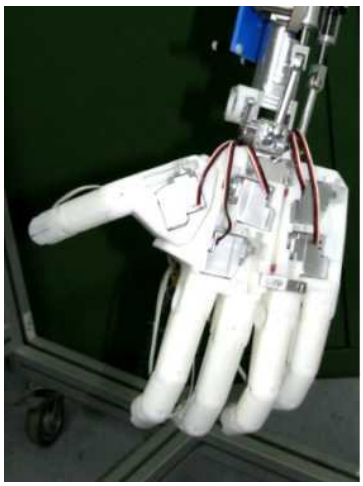
One of the first and very important requirement for the design of artificial hand is, that the visual solution should be as good as possible similar to the nature hand, and cannot be distinquished any differences between. Because of this, it should involve into considerations similar diamensions, and similar kinematic and dynamic characteristics, while the artificial hand should be covered with using very appropriate material.



a)



b)



c)

Figure 6. 3D model of the artificial hand:
a) Pro Engineer model,
b) 3D printed parts of the hand,
c) assembled prototype of the artificial hand

According to the presented above mechanical concept, the mechanism of the hand was designed in 3D CAD program ProEngineer, as it is shown in Fig. 6a. Based on stl files prodeced from this desin project, the 3D parts were printed using 3D printing machine Dimension Elite. Printed parts are presented in Fig. 6b. Then the prototype was assemled as it is shown in Fig. 6c.

As it was stated before, the artificial hand will be used to execute rather very simple taks: open and close, and a limited number of simple prehension. In typical such a solutions, functions of prehension could be selected from a single action patterns, similar to typical kind of prehension used most frequently in everyday activity of the men. The basic set of types of prehensions typically is created on the basis of prehension patterns in normal subjects. Most common and essential manipulations of such a gripper in relation to typical objects are then listed and described by lines of instructions of controlled program. So the set of prehensions contains: catching of the plastic bottle, catching a tennis ball, catching a rule, operating of drilling machine, etc., opening and closing of various locks, dialing telephones, carrying various objects, and the others. The fingers of the hand are opening when the cranks of appropriate motors are rotating to the back side, then the part of connecting rod connected with the crank pull the spring, which elastically pull the part of rod connecting to the up axis of drivong crank of first phalange of each finger. When all fingers are opening, the voluntary space inside the hand is enlarging. Then when the hand should catch selected object in the shape of ball or cylinder with adequate diameter, cranks of motors are driven in front direction, elastic rod is pushing elastically the driving crank of finger. When the finger is free, it is moving without any disfigurements, simultaneously with the motor crank. When the finger will touch of the external surface of an object, the spring will appropriately elastically deform generating the set of forces in points of contact, theses forces are dependent on resulted elastic deformations of appropriate springs.

Elements of connecting rods were made of duraluminium and bronze. Particular phalanges of the fingers and structural elements of the both driving system and the transmission one were equipeed with miniature ball bearings. The gripper driving system comprised miniature motors supplied with toothed gears, produced by the HITEC company. In each first axis of finger miniature magnetic sensor was applied. When motor shaft motion, the motor produce angular displacement measured by its oun sensor for the displacemnt of the first end of rod. Miniature sensor located in first joint of finger measure absolute angle of first phalange. When the finger interacts with environment, the difference of signals from motor and from magnetic sensor are measure deformations of elastic rod. Taking into account constant coefficient of stiffness of spring the sytem measure actual value of force transmitting throu elastic rod.

IV. FUNCTIONAL CHARACTERISTICS OF THE ARTIFICIAL HAND

Functional characteristics of the artificial hand show the sizes and masses of the objects, which it can grasp and manipulate. Those characteristics are developed taking into

consideration the driving system and at the assumed size limits of the manipulated objects. In the developed gripper design those characteristics depend on the corresponding characteristics of particular fingers. In the operation of a single finger the most important role plays the last phalange, in which the tactile sensor is located, and medium phalange, which is contacted with manipulated object. Because of applying special kinematic coupling between phalanges in each finger (the crossed five bar has been applied), last and medium phalanges during the revolution of the first phalange, rotates realised actual absolute angle, which value is an angle of previous phalange plus the value of angle of first phalange. By appropriate control, the hand can realize the motion in form outwards/inwards - opening/closing the hand. Each finger exerts the pressure upon the object in the direction perpendicular to the internal surface of phalange, the clamping force is produced by elastic force from rod driven by appropriate motor. The magnitude of that moment is limited by the maximum moment of the drive unit (motor with gear) and due to elastic deformation of elastic rod assumes the values between zero and maximal value of the motor moment:

$$M_u \in \langle 0, M_s \rangle$$

depending on the phalange loading and current value of the control signal. Magnitude of the force generated by the driving system depends on the angle of the phalange opening. Fig.7 presents kinematically coupled motion of the second and last phalange in the function of motion of the first phalange driven thtou rof from the motor and driving torque **T** generated by motor, then transmitted throu connecting rod as a force **F** which deform the spring with stiffness coeficient **k**, generated elastic deformation **x**.

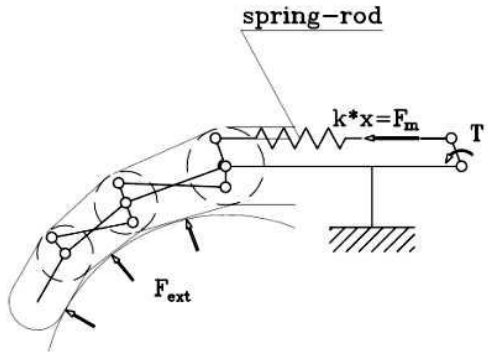


Figure 7. Schematic diagram of the driving system of the finger with kinematic coupling of phalanges motion.

The artificial hand is open, when fingers are straight. The ranges of fingers closing motion are similar to the human hand, and the maximum diameter of an object size is of about 100 mm. Fig. 8 present characteristic of the clamping force of the hand fingers in the course of manipulation.

The maximal value of the force varies in almost parabolic way (more precisely it is inverse-cosine) depending on the object size and for the maximal value of the fingers extension reaches the magnitude of 9N. In the case when the phalanges are covered with a rubber glove, the coefficient of friction, of which is about 0.35, the value of additional load carried due to friction and the shape reaches about 15N). It is worthwhile to note that the characteristics varies depending on the gripper opening and is approximately coincident with the characteristics of mass versus the object size. It is worthwhile to note that some manipulated object of complicated shapes can be grasped in a proper way, despite their relatively too high mass (weight).

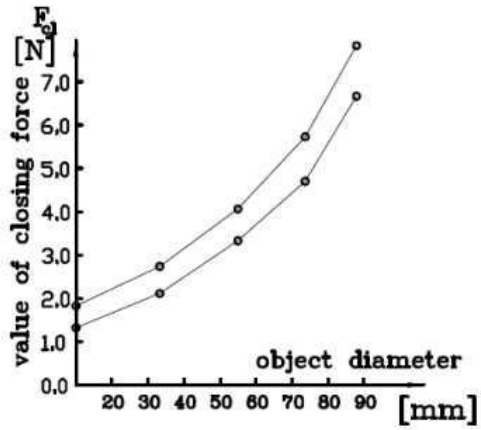


Figure 8. Dynamic characteristic of the clamping force of the designed hand

At the initial stage of the developed design of multi-fingered gripper of human hand form, it was assumed that a gripper consisting of three parts (subassemblies) - a palm divided into two parts with two finger each, and thumb (with additional revolution relative the longitudinal axis in its starting position) and two fingers with redundant DOF's pressing against the object, while two the other fingers will only assist the prehension. The fingers are actuated by means of the flexible elements that ensured automatic control of the pressure exerted by particular phalanges upon the object.

The driving systems are situated in the gripper body, while the phalanges are driven through connecting rods provided with additional systems of elastic springs. By adding flexibility to the fingers, we improve substantially the gripper functionality through rising its level of anthropomorphism. The fingers will be covered with a special flexible glove with embedded contact pressure force sensors and other measurement gauges. The gripper was designed for grasping an object by means of hooking clamping, grasping, supporting - stabilized by the other hand/gripper (like in a typical two-hand manipulation). Additionally, the design was supplied with the following means: force sensors, pressure sensors and contact sensors.

V. EXPERIMENTAL INVESTIGATIONS OF THE HAND

First studies of functional properties of the hand were investigated the process of catching simple objects like plastic bottle, small ball, machine for temperature gluing, and

the ruler. Examples from theses investigations are shown in Fig. 9.

Figure 9. Typical examples of prehension of investigated artificial hand.

Beside of examplws shown in Fig. 9, hand was mounted to arms of robot ROMAN. They can be used by robot for expression emotions. In Fig. 10 some typical nonwerbal expressions generated by system of artificial inteligence of robot are shown.

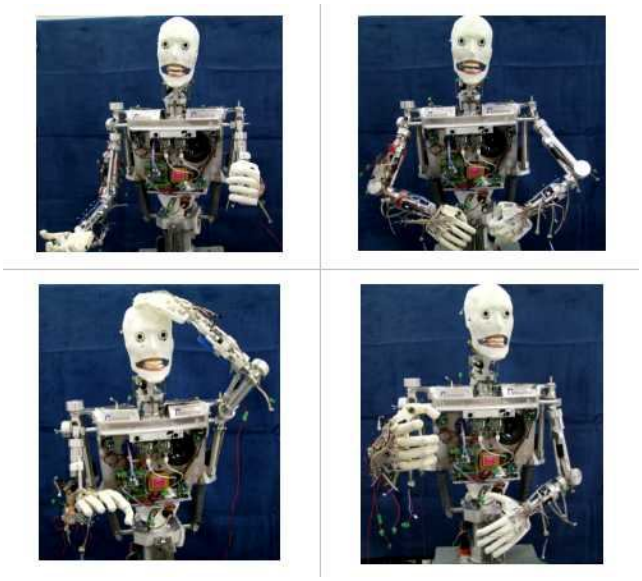


Figure 10. Examples of nonwerbal expressions generated by robot ROMAN. .

SUMMARY

This paper presents the result of the design and initial experimental investigations of the artificial hand with elastically deformed fingers. The novelty of presented work lies in the concept of the design in which motors power is limited while interacting forces between fingers and object are generated by elastic rods with sprins with appropriately improved dynamic gear ratio. Beside of this, the main aim of the experimentally aided design was to apply biological inspiration for robot design and his control. The proper design solutions should improve the robot performances and especially increase the dynamic properties while decrease the energy spendings. Applied iterative design procedure profited in the mass reduction, the geometry proportion and kinematic structure matches the human hands, the rapid prototyping method for production parts of the prototype caused that the obtained solution is relatively cheap and is characterized by high functionality. To ensure highest performance of the system, in the developed design, all mechanical, electric and electronic elements (necessary driving systems, gear and transmission systems, joints, sensors, etc.) are integrated with the mechanical system of a gripper . The mechanical part was designed using the ProEngineer system and then the initial simulations were performed using the ADAMS package, allowing for estimation of basics characteristics and their optimisation according to the assumed criteria. Presented results were validated by experiments.

REFERENCES

[1] Berns K., Asfour T., Dillmann R.: ARMAR – An Anthropomorfic Arm for Humanoid Robot, Proc. Int. Conf on Rbotics and Automation, An Arbor Mich. 1999, pp. 702-707,

[2] Kato I.: Development of WASEDA Robot – The Study of Biomechanisms at Kato Laboratory, 2nd edition, 1987,

[3] Mianowski K., Nazarczuk K., Słomkowski T.: Dynamic model for the selection of servomotors in serial-parallel manipulator, Int. Conf. CIM, Zeszyty Naukowe Polit. Śląskiej, seria Mechanika, z. 108, str. 261-268, Gliwice, 1992.

[4] Mianowski K.: Analiza właściwości chwytaka specjalnego przeznaczonego dla robota usługowego, mat. Kraj. Konf. TMM, Zielona Góra 2006, pp. 185-190. (in polish)

[5] Morecki A.: Basic of Robotics, Theory and Componenets of Robots and Manipulators, SpringerWienNewYork, 1999.

[6] Olędzki A.: Basics of TMM, WNT Warsaw, 1987. (in polish).

[7] Szynkiewicz W.: Planning of two-handed manipulation, in Postępy robotyki, by K. Tchon, WKŁ, Warszawa 2005, pp. 187-196. (in polish)

[8] Zieliński C., Szynkiewicz W., Mianowski K., Nazarczuk K.: Mechatronic design of open-structure multi-robot controllers, Mechatronics, 11(8), 2001, str. 987-1000.

[9] Zieliński C., Szynkiewicz W., Mianowski K., Rydzewski A., Winiarski T.: Efektory robota usługowego do dwuręcznej manipulacji z czuciem, proc. KKR 2006, (in polish)